Answer key: Lesson 2, Number Words and the Number Line

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill in the missing numbers in the boxes.

Answer: 10 and 12

The line goes 8, 9, 10, 11, 12, 13, 14. The empty boxes are 10 and 12.

Watch for: Writing 9 or 11 again (not counting on).; Leaving a box empty.

S2. Fill in the missing numbers.

Answer: 29; 39 and 41; 70

(a) 28, then 29, then 30. (b) 38, then 39, then 40, then 41: the line goes on after 40. (c) One before 71 is 70.

Watch for: Writing 310 for the number after 38 (the ones digit goes past 9).; Stopping at 40 and leaving the last box empty.

S3. Fill in the number just before and the number just after.

Answer: 34 and 36; 59 and 61; 78 and 80

(a) 34 and 36. (b) Back from 60 is 59, on from 60 is 61. (c) Back from 79 is 78. After 79 the ones are used up, so a new ten begins: 80.

Watch for: Writing 50 and 70 (a whole ten instead of one step).; Writing 710 for the number after 79.

S4. Write the numbers in digits.

Answer: 15, 50, 24

(a) fifteen is ten and 5: 15. (b) fifty is 5 tens: 50. (c) twenty is 20 and 4 more: 24.

Watch for: Writing 50 for fifteen, or 15 for fifty (teen and ty mixed up).; Writing 42 for twenty-four (digits the wrong way round).

S5. Write each number in words. Word bank: twenty, forty, sixty.

Answer: twenty, forty

20 is 2 tens: twenty. 40 is 4 tens: forty (no u).

Watch for: Writing fourty (forty has no u).; Copying the word for a different number, for example sixty for 40.

S6. Fill in the missing numbers. The numbers go up by 10 each time.

Answer: 30, 50 and 80

0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. The missing marks are 30, 50 and 80.

Watch for: Writing 21, 25 or 75 (counting on in ones instead of tens).; Leaving out the mark at 80.

S7. Fill in the blanks. Each number is between two tens.

Answer: 20 and 30; 60 and 70

(a) 23 is 2 tens and 3 more: between 20 and 30. (b) 68 is between 60 and 70.

Watch for: Writing 22 and 24 (the numbers just before and after, not the tens).; Writing 2 and 3 (digits) instead of 20 and 30.

S8. Ben counts: 57, 58, 59, 510, 511.

Answer: 60. After 9 the ones are used up, so a new ten begins. Ben kept counting ones.

After 59 the ones are used up, so we start a new ten: 6 tens and 0 ones, 60. Ben kept adding ones after the 9, so he wrote 510.

Watch for: Writing 510 (the ones digit goes past 9).; Circling Yes (not noticing 510 follows 59).

Think harder (page 3)

T1. Which ten is nearer? Fill in the blanks.

Answer: 50; 90

(a) 47 to 50 is 3 hops, 47 to 40 is 7 hops: 50. (b) 89 to 90 is 1 hop, 89 to 80 is 9 hops: 90.

Watch for: Choosing the ten with the same tens digit (40 for 47) without counting the hops.; Counting the hops from the wrong ten.

T2. How many numbers are there between 28 and 35? (Do not count 28 and 35.)

Answer: 6 (29, 30, 31, 32, 33, 34)

29, 30, 31, 32, 33, 34. That is 6 numbers. The hops from 28 to 35 are 7, but the numbers in the middle are one fewer.

Watch for: Answering 7 (counting the hops, or counting 35).; Answering 8 (counting 28 and 35 as well).

T3. My number is between 50 and 60. It is nearer to 60 than to 50. The number before it is halfway between 50 and 60. What number is it?

Answer: 56

Between 50 and 60 and nearer to 60: 56, 57, 58 or 59. The halfway number is 55. One after 55 is 56, so clue 3 fits only 56.

Watch for: Answering 55 (that is halfway, not nearer to 60).; Answering 54 (one before 55: clue 3 says one before ME is halfway).

T4. Mei's frog is on 90. It hops 2 numbers at a time.

Answer: 96 and 98; No (it lands on 98, then 100)

90, 92, 94, 96, 98, 100. The frog lands on 96 and 98. It hops over 99 (from 98 to 100), so it does not land on 99.

Watch for: Writing 95 and 96 (hopping 1 at a time).; Circling Yes for 99 (counting every number, not every second number).

Go further (page 4)

G1. Complete each line. The jumps are equal.

Answer: (a) 10, 20, jump 10. (b) 15, 20, 25, jump 5. (c) 53, 56, 59, jump 3.

(a) 3 jumps from 0 to 30: 10 each. (b) 4 jumps from 10 to 30, which is 20 altogether: 5 each. (c) 4 jumps from 50 to 62, which is 12 altogether: 3 each.

Watch for: Always using a jump of 10.; Counting the boxes (3) as the jumps (4).

G2. Mei's frog is on 28. Ali's frog is on 52. They hop towards each other, 1 number at a time, at the same time. On which number do they meet?

Answer: 40

From 28 to 52 is 24 hops (28 to 30 is 2, 30 to 50 is 20, 50 to 52 is 2). Each frog makes half: 12 hops. 28 and 12 more is 40. 52 and 12 back is 40. They meet on 40.

Watch for: Choosing 38 or 50, one frog's own number, or adding 28 and 52.; Counting the hops wrongly across the tens.

G3. Make a puzzle. The jumps are equal. Write numbers in 2 boxes that are not next to each other. Leave 3 boxes empty. Give it to a grown-up.

Answer: Any correct line, for example 15, ___, ___, ___, 35 (jump 5: 20, 25, 30).

Example: jump 5: 15, 20, 25, 30, 35. Hide 20, 25 and 30. The grown-up counts the 4 jumps from 15 to 35, which is 20 altogether, so the jump is 5, and the boxes are 20, 25 and 30. Check: the same jump between every two boxes.

Watch for: Making jumps that are not equal.; Giving two numbers that do not fit one equal jump.

Investigation (page 5): Halfway

41, 42, 43 and 44 are nearer to 40. 46, 47, 48 and 49 are nearer to 50. 45 is exactly halfway: 5 hops each way. Rule: ones digit 1 to 4, the lower ten is nearer; ones digit 6 to 9, the higher ten is nearer; ones digit 5, halfway. It works between every two tens, for example 61 to 69.

Look for: A child who tests several numbers by counting hops, notices 45 is special, and says why the ones digit decides it. A child who finds the same pattern between 60 and 70 has generalised.

Exit ticket (page 5)

E1. Write the number in digits.

Answer: 73

Seventy is 70, and 3 more makes 73.

Watch for: Writing 37.; Writing 703.

E2. Fill in the blanks. 76 is between two tens.

Answer: 70 and 80

76 is 7 tens and 6 more, so it is past 70 and not yet 80.

Watch for: Writing 75 and 77 (just before and after, not the tens).; Writing 7 and 8.; Writing 60 and 70.

E3. The frog is on 99. It hops 1 number forward. Where does it land?

Answer: 100

After 99 comes 100, one hundred: ten tens.

Watch for: Writing 910 or 9100.; Writing 90.

Sneak-peek puzzles (not in the workbook)

1. 20 and 23
2. 40. From 38 to 40 is 2 hops. From 38 to 30 is 8 hops.